

TEST BANK

CHAPTER 5

SHALLOW-WATER SUBTIDAL BENTHIC ASSOCIATIONS

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A and then place the letter for the selected choice from column B in the space provided to the left of the corresponding item in column A.

[Note to instructors: Here, the choices are correctly matched and thus should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. deposit-feeding clams	_____→	A. sediment destabilizers
___ 2. interference competition	_____→	B. trophic group amensalism
___ 3. weasel predators	_____→	C. burrowing predators
___ 4. gametophyte	_____→	D. sexual microscopic stage
___ 5. pneumatocyst	_____→	E. float device
___ 6. macrofauna	_____→	F. organisms > 0.5 mm long
___ 7. meiofauna	_____→	G. organisms 0.5–0.062 mm long
___ 8. microfauna	_____→	H. organisms < 0.062 mm long
___ 9. cnidarian	_____→	I. sea pen
___ 10. echinoderm	_____→	J. sea urchin

Multiple Choice

Circle the choice that best completes or answers the given questions.

[N.B.: The correct answer to each of the questions is indicated by an asterisk. For most of the questions, the letter “a” is the correct choice; reorder the choices as you see fit.]

1. A parallel bottom community refers to which of the following?

a. Similar, but not identical, organisms inhabiting geographically separate regions *

b. The same community with the same species in two geographically separate regions

c. Similar bottom and sediment types with very different organisms

d. Similar organisms, but in regions with very different bottom and sediment types

2. Exclusion of one species from a shallow sub-tidal region because of the general activities of another species (not including predation) is called which one of the following?

a. Competitive interference *

b. Competitive exploitation

c. Disturbance

d. Foraging

e. Grazing

132 Test Bank for Marine Biology

3. A member of the infauna that, by its presence in the sediments, helps to hold the sediments in place is called a _____.
 - a. sediment stabilizer *
 - b. bioturbator
 - c. parallel bottom community
 - d. keystone predator
4. Which one of the following scientists invented a device to collect samples of the infauna?
 - a. Petersen *
 - b. Thorson
 - c. Vance
 - d. Dayton
 - e. Darwin
5. The amount of light penetrating the inshore water is usually ____ that penetrating the open ocean.
 - a. less than *
 - b. greater than
 - c. equal to
6. The ____ lives on or upon the surface of the marine bottom.
 - a. epifauna *
 - b. infauna
 - c. meiofauna
 - d. macrofauna
 - e. microfauna
7. The zone between the lowest low-tide line and the continental shelf at 200 m is the ____ zone.
 - a. sublittoral *
 - b. neritic
 - c. pelagic
 - d. littoral
 - e. planktonic
8. Animals that burrow through the substrata of the benthos make up the _____.
 - a. infauna *
 - b. epifauna
 - c. pelagic fauna
 - d. neritic fauna
 - e. None of the above
9. The concept of parallel bottom communities was developed chiefly from the work of which one of the following researchers?
 - a. Thorson *
 - b. Petersen
 - c. Cousteau
 - d. Darwin
 - e. Nybakken
10. Animals that cause the sediments to shift, become resuspended, or otherwise change are called _____.
 - a. bioturbators *
 - b. *r*-selected
 - c. *K*-selected
 - d. sediment stabilizers
 - e. parallel bottom communities
11. Animals that are exposed at the surface of the sediment and eat organisms from the surface or just above it are called ____ predators.
 - a. surface *
 - b. burrowing
 - c. weasel
 - d. digging
12. Animals that move down tubes or channels to attack their prey are called ____ predators.
 - a. weasel *
 - b. surface
 - c. browser
 - d. digging
 - e. More than one of the above
13. Animals that excavate holes to get to their food are called ____ predators.
 - a. digging *
 - b. burrowing
 - c. weasel
 - d. browser
 - e. surface
14. The presence, in the same general area, of patches comprising different groups of organisms is referred to as _____.
 - a. alternate stable states *
 - b. trophic group amensalism
 - c. keystone species
 - d. parallel bottom communities
 - e. Thorson's rule

15. Kelp beds and forests form throughout the world in ____ with ____ bottoms.
 - a. cold temperate regions, hard *
 - b. cold temperate regions, soft
 - c. warm temperate regions, hard
 - d. warm temperate regions, soft
16. The exclusion of organisms by modification of the environment by another organism is called _____.
 - a. trophic group amensalism *
 - b. mutualism
 - c. parasitism
 - d. competitive interference
17. ____ are those animals that live within the sediments at all times, burrowing through them to find their prey.
 - a. Infaunal predators *
 - b. Digging predators
 - c. Weasel predators
 - d. Burrowing predators
18. The lower limit of the algal-dominated rocky infralittoral zone of the cold temperate region extends to about ____ m.
 - a. 25 *
 - b. 250
 - c. 1
 - d. 100
19. A consequence of wasting disease on the Scandinavian eelgrass beds is ____ in the intertidal region.
 - a. erosion *
 - b. deposition of sediments
 - c. an increase in plant production
 - d. an increase in the extent of the *Zostera* meadows
20. The North Pacific sea pen is a member of which of the following animal phyla?
 - a. Echinodermata *
 - b. Chordata
 - c. Mollusca
 - d. Vertebrata
 - e. Arthropoda

True/False

Mark each of the given statements with the letter *T* if it is true and *F* if it is false.

[Instructors: In some cases, a few options are included for the same question.]

1. [True] A functional group is all the species of all taxa in a particular geographic area that use the resources in the environment and affect the environment in similar ways.
2. [True] The zone between the lowest low-tide line and the continental shelf at 200 m is the sublittoral zone.
3. [False] The zone between the lowest low-tide line and the continental shelf at 200 m is the pelagic zone.
4. [False] When kelps form a floating surface canopy, the association is known as a kelp forest.
5. [True] Sea-ice diatoms, on occasion, achieve population levels at which they actually color the bottom of ice flows.
6. [False] The macroscopic stage of kelp is the gametophyte stage.
7. [True] Kelp forests can comprise both an overstory of kelp and an understory of sea grass.
8. [False] The characteristic feeding position for sand dollars is flat on their ventral surface.
9. [True] While kelps are very productive, it has been estimated that only about 10% of their net production enters the food web through direct grazing.
10. [True] Most of the net production of kelps enters the food web through detritus or DOM.
11. [True] Of central importance to the Antarctic seas is the krill, *Euphausia superba*.
12. [True] In the Antarctic, hard-bottom communities are more common than soft-bottom communities.
13. [True] The presence, in the same general area, of patches comprising different groups of organisms is referred to as alternate stable states.

134 Test Bank for Marine Biology

14. [True] Keystone species are often predators that are capable of structuring the community.
15. [True] Petersen was a Danish researcher who invented a device to collect samples of the infauna.
16. [False] Cage—or exclusion—experiments have provided little information regarding the importance of predators in subtidal benthic habitats.
17. [True] Animals that cause the sediments to shift, become resuspended, or otherwise change are called bioturbators.
18. [True] Animals that cause the sediments to shift, become resuspended, or otherwise change are called sediment destabilizers.
19. [False] Psychrophilic bacteria are warm-water-loving bacteria.
20. [True] In the winter, production by psychrophilic bacteria may be greater than algal primary production.
21. [True] Kelp beds and forests tend to be absent from a circumglobal zone of about 5°S to 20°N latitude.
22. [False] One of the major differences between the Arctic Ocean Basin and the Antarctic region is the presence of an extensive shallow shelf in the Antarctic.
23. [True] Infaunal animals distribute themselves vertically in the benthos.
24. [False] Anchor ice is a sheet of ice that attaches to the sediment from the subice platelet layer to well over 100 m in depth.
25. [False] In microbial food webs, DOM is unimportant.
26. [False] There is no significant difference between the relative abundances of metazoans in the sea ice of the Arctic and the Antarctic.
27. [True] In the Arctic Ocean, there is no dominating euphausiid shrimp in the pelagic zone.
28. [False] Kelp beds are permanent features of the cold temperate parts of the world's oceans that possess hard subtidal substrates.
29. [True] Sponges, soft corals, and pynogonoids are animals found in unvegetated sediments at the polar latitudes. These organisms are less common in similar habitats of temperate latitudes.
30. [False] Burrowing deposit feeders are most abundant in sand, while in mud habitats suspension feeding dominates.

Short Answer (Fill in the Blank)

1. Benthic organisms can become entrapped in anchor ice. When this ice breaks away from the benthos, it carries the organisms with it.
2. Organisms that cause the sediments to shift, become resuspended, or erode are known as bioturbators.
3. Herbivores in unvegetated sedimentary environments feed as suspension feeders or deposit feeders.
4. The dominant genus of kelp in the North Atlantic Ocean is Laminaria.
5. The exclusion of a species that occupies one trophic level by another species that occupies a different trophic level is called trophic group amensalism.
6. The dominant genus of kelp along the coast of the Pacific Northwest is Macrocystis.
7. Kelp blades possess pneumatocysts—or floats—that help keep the blades afloat at the surface.
8. Kelps are attached to the substratum by a structure called a holdfast.
9. The dominant plant form in a kelp forest is the sporophyte stage.
10. Cages that exclude epifaunal predators from soft sediments usually result in increased numbers of infaunal organisms.
11. The most common type of animal on subtidal rocks is colonial encrusting species which are capable of spreading over the surface by asexual reproduction.
12. Thorson was the researcher who established the concept of parallel bottom communities.

Essay Discussion Questions

Develop complete answers to these questions.

1. What factors tend to allow for the high productivity of the shallow-water regions? What influence does this have on animal production?
2. What is the Petersen grab sampler? Why was it invented? What does it do?
3. What is the concept of parallel bottom communities? What does this concept mean ecologically? What does it imply (perhaps) to evolutionary ecological theory?
4. Discuss the terrestrial counterpart to parallel bottom communities.
5. Provide an argument to support the following statement: The fraternity–sorority system in colleges and universities is analogous to the marine concept of parallel bottom communities.
6. Why does the concept of parallel bottom communities seem to fit in temperate, but not tropical, waters? Discuss.
7. Does the evidence of (temperate) parallel bottom communities fit a community model or a continuum model? Provide examples to support your essay.
8. Describe the concept developed by Rhoads and Young (1970) of trophic group amensalism.
9. Describe the concept of competitive interference (interference competition). Compare it with exploitative competition. Discuss.
10. An analogous concept to competitive interference, but in the opposite direction, is called facilitation. What does it mean in terms of the infauna? Describe a scenario involving two or three hypothetical shallow-water benthic species in which facilitation takes place.
11. How did the cage-exclusion experiments of Virnstein (1977) and others demonstrate that predation is an important factor in structuring the infauna community?
12. How have cage experiments helped us to understand the importance of predation in structuring the sublittoral benthos? Discuss, and cite one example.
13. What is a keystone species? What is a keystone predator? Name some keystone species or predators in terrestrial ecosystems and in aquatic ecosystems.
14. In regions where there is abundant food for infaunal suspension feeders, how is the habitat structured (distribution of species) so as to pack more species into a given area?
15. What does the term “commensalism” mean? Give an example of an infaunal commensal relationship.
16. Do you see any common geographic factors in the distribution of the major kelp beds of the world? What are the major kelp genera of the world?
17. What sorts of growth rates do kelp plants have? Compare this growth with the productivity of terrestrial systems and phytoplankton.
18. How is a healthy population of sea otters linked to a healthy kelp-bed community? Discuss.
19. Compare the productivity of kelps and seagrasses.
20. Of what ecological importance are seagrasses in the inshore area?
21. Of what ecological importance are kelps in the inshore area?
22. What is a sea pen? To what phylum does it belong? How does it feed? What is unusual about large sea-pen communities? Discuss.
23. What is anchor ice? How do anchor and annual sea ice affect the bottom community in the Arctic? Compare these features in the marine environment with those of a freshwater lake that freezes over each winter.
24. Discuss the ecological and survival advantages of planktonic larval stages for benthic animals living in shallow water.
25. What does the term “grain size” mean with respect to the ecology of the benthos? (*Hint*: What does grain size mean in photography?)

136 Test Bank for Marine Biology

26. Design an organism that, under conditions of fine, organic rich sediments, is a sediment destabilizer, while under conditions of coarse sandy sediments is a sediment stabilizer.
27. How are alternative stable states (Sutherland, 1974) like the subclimax sere present in terrestrial ecosystems?
28. What common features do all sediment destabilizers possess?
29. What common features do all sediment stabilizers possess?
30. Of what importance are sediment stabilizers to the benthos? Discuss.
31. Of what importance are sediment destabilizers to the benthos? Discuss.
32. Distinguish between the concepts of direct and indirect competition. Give an example of each from the marine environment.
33. Distinguish between the concepts of interference and exploitative competition. Give an example of each from the marine environment.
34. Describe the concept of trophic group amensalism, and cite an example of its documentation in a marine situation.
35. Describe the concept of trophic group amensalism, and cite an example of its documentation in a terrestrial situation.
36. How does the sea cucumber *Molpadia oolitica* function as a bioturbator and as a trophic group amensal? Discuss.
37. Examine the ecological significance of bat-ray holes.
38. What is a sediment stabilizer organism? Provide examples.
39. What is a sediment destabilizer organism? Provide examples.
40. How could competition for space in the sediments lead to the evolution of commensal relationships within the infauna? Discuss, supporting your argument by citing examples.
41. How does the creation of stable sediment cones surrounded by unstable depressions by the sea cucumber *Molpadia* both help stabilize the community and exclude certain suspension feeders?
42. Although the terms are often used interchangeably, what are the structural differences between a kelp bed and a kelp forest? How do these differences affect the rest of the community?
43. What problems would kelps encounter if they grew from their apex (top) rather than from their base?
44. What problems would sea grasses encounter if they grew from their apex (top) rather than from their base?
45. Investigate the ecological significance of the fact that most of the net production of kelps enters the food web through detritus or DOM.
46. Discuss how sea urchins are important to the kelp beds.
47. Explain the differences between direct and indirect competition, and provide examples of each.
48. Discuss the experimental technique of caging. Why is it done, and what might it demonstrate?
49. One difference between temperate seagrass beds and seagrass beds of tropical waters is that the temperate beds accumulate sediments that are from sources outside the beds, whereas tropical seagrass beds accumulate sediments from within. Discuss the ecological significance of this fact.
50. How might seagrass provide a refuge for bivalves from predatory gastropods? Discuss.
51. How is seagrass detritus channeled out of seagrass beds to other shallow-water systems?
52. What important roles do seagrass beds serve in inshore regions? Discuss at least two such functions.
53. Explain how seagrass beds are nursery grounds for numerous species.
54. What are the consequences of wasting disease for a seagrass community and for the soft-bottom community?
55. Discuss the evidence which indicates that shallow-water benthic communities are not completely stable over long periods of time.

56. Compare and contrast the Arctic and Antarctic in terms of the benthic environment.
57. How does anchor ice help structure the shallow-water benthic community of the poles?
58. Present an argument that would explain why the fauna of the Arctic Ocean is impoverished and has a low rate of endemism when compared with the fauna of the Antarctic region.
59. How does the energy flow from a kelp bed (forest) into its food web, and how is that energy flow similar to that from a forest into its food web? Discuss.
60. How does a seagrass bed lead to a reduced oxygen concentration in the sediments? Discuss.
61. Compare and contrast productivity levels in kelp and seagrass beds.
62. How are seagrasses stabilizing organisms?
63. What roles do seagrasses serve in shallow inshore waters?
64. How does seagrass disease influence inshore communities? Discuss.
65. The euphausiid shrimp *Euphausia superba* holds a central position in the trophic dynamics of Antarctic seas. Discuss.
66. Explain the hypothesis of alternate stable states.
67. How do kelps and sea grasses differ in their ability to take up and recycle nutrients?
68. Compare the vertical structure of seagrass and of kelp beds. Remember to consider the entire plant.
69. Compare and contrast Antarctic hard- and soft-bottom communities.
70. Challenge Question: As the tide moves up into an estuarine-river system, which side of the estuary will receive the salt water first? (*Hint*: Analyze this problem in terms of the Coriolis effect.)
71. Challenge Question: Nybakken reports in his textbook that Antarctic organisms show more endemism than organisms of the Arctic. Speculate as to why this is the case.
72. Challenge Question: It is said that the benthos is usually a mosaic of patches of different ages, each with a different set of species. Discuss why this is the case.

